

Evaluating the Robustness of Pathology Foundation Models to a Feature Space Adversarial Attack

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Abstract

Pathology Foundation Models (PFMs) are increasingly used as frozen feature extractors for downstream tasks in digital pathology. Their robustness to adversarial perturbations is therefore an important aspect to investigate. This study evaluates the robustness of three PFMs, UNI, CONCH, and Virchow, against the task-agnostic feature-space adversarial attack proposed by Pulfer et al. on a 5-class subset of the TCGA-UT dataset. Under white-box conditions, the feature representations extracted by all three PFMs show progressively greater disruption as the attack strength increases, as reflected in larger shifts measured through cosine similarity and greater degradation of cancer-type discriminability assessed through linear probing. Under black-box conditions, the same attack has a substantially smaller effect on the evaluated models, although this should not be taken as general evidence of robustness to black-box attacks. As a defensive strategy, Gaussian smoothing partially mitigates the effects of the adversarial perturbations, but its effectiveness is limited by the accompanying loss of morphological details. Given the role of PFMs in downstream AI models, the observed vulnerability of their feature representations raises concerns about the robustness of clinical applications built upon them.

1 Introduction

The introduction of digital slide scanners has enabled tissue samples to be digitized into high-resolution, gigapixel-scale whole-slide images (WSIs). The application of artificial intelligence (AI) to these data has given rise to AI-based digital pathology, a field that has shown promising results across a wide range of tasks, including cancer detection and subtyping, biomarker profiling, prognosis estimation and recurrence risk assessment [11].

Because WSIs are extremely large, a typical processing workflow first identifies the tissue region and divides it into smaller patches. Features are then extracted independently from each patch, representing the WSI as a bag of patch-level features. These features are subsequently aggregated, for instance using attention-based multiple instance learning (ABMIL) [3] or other aggregation approaches, into a single slide-level representation, which is then used for the specific downstream task of interest.

Many downstream AI models rely on Pathology Foundation Models (PFMs) as frozen feature extractors, based on the premise that PFMs, pre-trained using self-supervised learning on large-scale histopathology datasets, can extract general-purpose representations from tissue patches. Multiple studies have evaluated PFMs from a performance perspective [1][10][8] and some have specifically examined their robustness to distribution shifts, such as those arising from differences in tissue preparation and scanning workflows across medical centers [4][5]. However, comparatively less attention has been given to their robustness against adversarial perturbations, despite the particular relevance of this aspect in safety-critical domains such as healthcare.

A recent study by Wang et al. [15] has investigated the vulnerability of PFMs to adversarial perturbations, proposing a Universal and Transferable Adversarial Perturbation (UTAP) attack. This study aims to evaluate the robustness of the feature representations learned by three PFMs - UNI [2], CONCH [7] and Virchow [14] - using an alternative adversarial attack, specifically the task-agnostic feature-space adversarial attack proposed by Pulfer et al. [13]. The PFMs are first subjected to adversarial perturbations specifically optimised for each model in a white-box setting, in which the attacker has full access to the target model and its gradients. They are subsequently evaluated using adversarial images generated on a different model, representing a black-box scenario in which the attacker has no access to the target model. The impact of the adversarial attack is examined through an analysis of the feature space, including UMAP visualization and quantification of embedding shifts, as well as through the evaluation of representation separability based on linear probing. Gaussian smoothing is then evaluated as a potential defense. Finally, the ethical implications of PFM vulnerability to adversarial attacks are discussed.

2 Dataset

The TCGA-UT dataset [6] is a publicly available collection of over 1.6 million image patches of hematoxylin & eosin (H&E)-stained histological samples from 31 human cancers. The image patches are 256×256 pixels in size, with a resolution of $0.5 \mu\text{m}/\text{pixel}$, corresponding to a physical tissue area of $128 \times 128 \mu\text{m}$. Due to computational constraints, the experimental pipeline is restricted to a subset of five cancer types: bladder urothelial carcinoma, colon-rectum adenocarcinoma, prostate adenocarcinoma, skin cutaneous melanoma and thyroid carcinoma. The selected subset comprises a total of 51,250 image patches. The official dataset splits are adopted, comprising a training set, a validation set and a test set, with 70%, 15% and 15% of the data, respectively. These splits are defined at the patient level, ensuring that no patient contributes patches to more than one split, thus preventing data leakage. Table 1 reports the number of patches per class and data split.

Class	Train	Valid	Test	Total
Bladder Urothelial Carcinoma	6,990	1,500	1,500	9,990
Colon-Rectum Adenocarcinoma	7,020	1,510	1,500	10,030
Prostate Adenocarcinoma	6,870	1,470	1,470	9,810
Skin Cutaneous Melanoma	7,040	1,510	1,510	10,060
Thyroid Carcinoma	7,950	1,710	1,700	11,360
Total	35,870	7,700	7,680	51,250

Table 1: Number of patches per class and data split.

3 Pathology Foundation Models

Three state-of-the-art pathology foundation models are evaluated: two vision-only models, UNI [2] and Virchow [14], and a vision-language model, CONCH [7], for which only the vision encoder is employed. Although they all leverage a Vision Transformer (ViT) backbone, they differ in their specific model architectures, large-scale pre-training datasets, pre-training strategies and embedding dimensionalities.

UNI adopts a ViT-Large architecture (ViT-L/16), with the [CLS] token used as the feature embedding, resulting in a 1024-dimensional representation. The model is pre-trained through self-supervised learning based on DINOv2. Its pre-training dataset, Mass-100K, comprises more than 100 million tissue patches sampled from 100,426 diagnostic H&E WSIs, collected from Massachusetts General Hospital (MGH), Brigham and Women’s Hospital (BWH) and the Genotype-Tissue Expression (GTEx) consortium.

Virchow employs a ViT-Huge architecture (ViT-H/14), with the feature embedding obtained by concatenating the [CLS] token with the average pool of the patch tokens, resulting in a 2560-dimensional representation. The model is pre-trained through self-supervised learning based on DINOv2 on approximately 1.5 million H&E-stained WSIs from Memorial Sloan Kettering Cancer Center (MSKCC).

CONCH uses a ViT-Base architecture (ViT-B/16) as its vision encoder. The image encoder is first pre-trained using the self-supervised iBOT algorithm on an in-house dataset of 16 million image tiles, sampled and extracted from the tissue regions of 21,442 WSIs, then fine-tuned together with a text encoder/decoder using the CoCa loss (image-text contrastive loss and captioning loss) on 1.17 million human histopathology image-caption pairs from publicly available PubMed Central Open Access (PMC-OA) and internally curated sources. The feature embedding of the vision encoder is obtained from the attentional pooler used for image–text alignment, resulting in a 512-dimensional embedding.

Notably, none of these foundation models were pre-trained on TCGA data, thereby preventing data leakage between the dataset used in this study and the data used for PFM pre-training.

4 Adversarial Attack

The adversarial robustness of the pathology foundation models is evaluated against a task-agnostic adversarial attack based on the framework proposed by Pulfer et al. [13] and its official implementation [12]. By modifying the input image with visually imperceptible noise, the attack aims to maximally disrupt the feature representation obtained by the foundation model, shifting the feature embedding of the adversarial image far from that of the original image.

Formally, let $f_\theta : \mathcal{X} \rightarrow \mathcal{Z}$ denote a pathology foundation model that maps an input image

$x_0 \in \mathcal{X}$ to its feature embedding $z_0 = f_\theta(x_0) \in \mathcal{Z}$. The attack aims to generate an adversarial image $x_{\text{adv}} = x_0 + \delta$ by optimizing a perturbation δ that minimizes the feature-space loss $\mathcal{L}_z(z_0, z_{\text{adv}})$, defined as the cosine similarity between the original and adversarial embeddings:

$$\mathcal{L}_z(z_0, z_{\text{adv}}) = \text{CosSim}(z_{\text{adv}}, z_0) = \frac{z_{\text{adv}} \cdot z_0}{\|z_{\text{adv}}\|_2 \|z_0\|_2} \quad (1)$$

where $z_{\text{adv}} = f_\theta(x_{\text{adv}}) \in \mathcal{Z}$ is the feature embedding of the adversarial image.

Adversarial perturbations are generated using Projected Gradient Descent (PGD) with a step size $\alpha = 5 \times 10^{-4}$ for a maximum of 350 iterations. Following Pulfer et al., the perturbation is constrained by a maximum per-pixel budget $\|\delta\|_\infty \leq \epsilon$, with ϵ set to $8/255$. This constraint is kept fixed across all experiments, while the attack strength is controlled by targeting a specific Peak Signal-to-Noise Ratio (PSNR), which constrains the overall image-space distortion, defined as:

$$\text{PSNR} = 10 \log_{10} \left(\frac{1}{\text{MSE}} \right), \quad \text{MSE} = \frac{1}{3HW} \|x_0 - x_{\text{adv}}\|_F^2 \quad (2)$$

where $x_0, x_{\text{adv}} \in [0, 1]^{H \times W \times 3}$ are the original and adversarial images, respectively. The PSNR values tested are 45 dB, 40 dB and 35 dB, with lower PSNR values corresponding to greater overall image distortion.

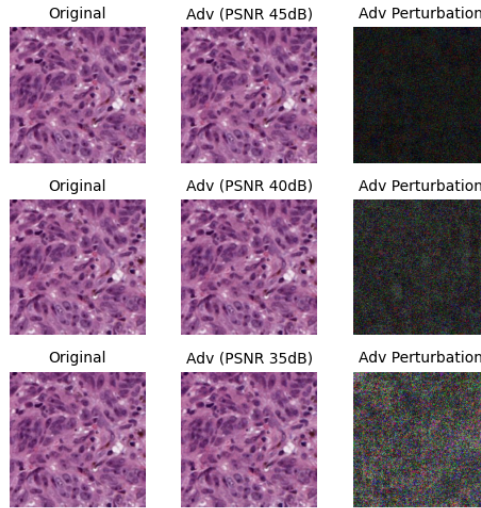


Figure 1: Adversarial images and corresponding perturbations at different PSNR values..

Figure 1 shows the original and adversarial images, together with the corresponding adversarial perturbations, for each of the three PSNR values tested. The perturbations are visualized as the absolute pixel-wise differences between the adversarial and original images, amplified for clarity:

$$\delta_{\text{vis}} = \text{clip}(|x_{\text{adv}} - x_0| \cdot s, 0, 1)$$

where $s = 20$ is the scaling factor used to highlight the otherwise imperceptible changes introduced by the attack. The visualized perturbations become progressively more pronounced from 45 dB to 35 dB, reflecting the increasing image distortion associated with lower PSNR values.

5 Gaussian Smoothing as Defense

Given the use of the PFM as a frozen feature extractor, a pre-processing-based defense is considered rather than strategies that require modifying the model weights, such as adversarial training.

Since adversarial perturbations primarily manifest as high-frequency spatial noise, low-pass spatial filtering is evaluated as a pre-processing defensive strategy to attenuate adversarial noise prior to feature extraction. Specifically, Gaussian filtering is employed and standard deviation values $\sigma \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0\}$ are tested. The kernel size is chosen based on the σ value using the rule-of-thumb that sets the kernel dimensions to $(2k + 1) \times (2k + 1)$ where $k = \lceil 3\sigma \rceil$, given that the $[-3\sigma, +3\sigma]$ interval captures approximately 99.7% of the area under the Gaussian function.

The decision to evaluate a range of σ values is driven by the fact that a higher σ yields stronger smoothing; evaluating this range enables an analysis of the trade-off between effective adversarial noise attenuation and the side effect of filtering out fine-grained morphological details of the histopathology image.

As illustrated in Figure 2, higher values of σ produce stronger blurring on both original and adversarial images, causing a progressive loss of morphological detail.

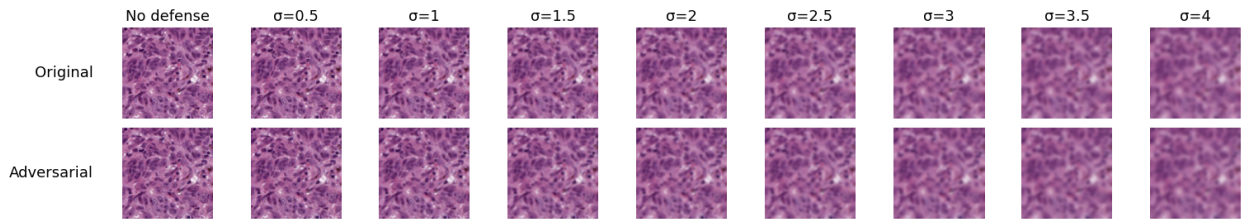


Figure 2: Effect of Gaussian filtering with increasing σ on original and adversarial images.

6 Experimental Pipeline

This section summarises how the dataset, the pathology foundation models, the adversarial attack and the defensive strategy are integrated into the overall experimental pipeline.

6.1 Feature Extraction from Clean and Adversarial Images With and Without Defense

Image Preprocessing Input images are preprocessed following the official protocols specified by the authors of each pathology foundation model. Specifically, each 256×256 patch is resized and center-cropped to 224×224 pixels, then normalized using model-specific mean and standard deviation values.

Feature Extraction from Clean Images Feature extraction is first performed on the clean, unperturbed images from the training, validation and test sets using each pathology foundation model. The resulting feature embeddings are saved to disk as HDF5 files for subsequent analysis.

Adversarial Image Generation and Feature Extraction Adversarial images are generated from the test set for each pathology foundation model and PSNR value, and saved as PyTorch tensor files. Feature extraction is then performed under two distinct scenarios, with all resulting embeddings saved as HDF5 files:

- **White-box scenario:** Features are extracted from the adversarial images using the same model used to generate them (target model = source model).

- **Black-box scenario:** Features are extracted from the adversarial images using a model different from the source model (target model $\neq$ source model), thereby evaluating the cross-model transferability of adversarial perturbations.

Defensive Strategy and Feature Extraction To assess the impact of the defensive strategy in the white-box scenario, Gaussian smoothing is applied separately to both clean and adversarial images for each σ value prior to feature extraction, with the resulting features saved to disk as HDF5 files.

6.2 Evaluation Protocol

Feature Space Analysis The impact of the white-box adversarial attack is analysed using Uniform Manifold Approximation and Projection (UMAP) [9] to qualitatively visualise the feature space of pathology foundation models before and after the attack, while cosine similarity between clean and adversarial features is computed to quantify the embedding shift.

Representation Separability Assessment Linear probing is used to assess the effect of adversarial perturbations on cancer-type discriminability and to evaluate the effectiveness of the defensive strategy.

- **Training and Baseline Performance:** For each pathology foundation model, a linear probe consisting of a linear classification layer is trained on the corresponding clean training embeddings using the Adam optimizer with a learning rate of 5×10^{-4} for a maximum of 30 epochs. The linear probe checkpoint achieving the highest balanced accuracy on the clean validation set is selected and subsequently evaluated on the clean test set. The balanced accuracy achieved on the clean test set serves as the baseline performance.
- **Attack Evaluation:** The trained linear probe is evaluated on the adversarial embeddings corresponding to each PSNR value under both white-box and black-box scenarios. Performance is measured using balanced accuracy.
- **Defense Evaluation:** The trained linear probe is evaluated on the embeddings generated from smoothed clean and smoothed adversarial images under the white-box scenario. Performance is measured using balanced accuracy for each tested σ value to quantify both performance recovery on the adversarial set and performance degradation on the clean set.

7 Results

7.1 Feature Space

The UMAP visualizations in Figure 3, projecting the feature spaces of all PFMs into two dimensions, show that, in the absence of attacks, both the training and test embeddings form largely well-separated clusters corresponding to the five cancer types. In contrast, following the application of the attack to the test images under white-box conditions, the class-related organization observed in the clean setting is progressively lost across all PFMs as PSNR decreases from 45 dB to 35 dB.

Figure 4 shows the cosine similarity between the embeddings of clean and adversarial images across the different PSNR values.

Cosine similarity ranges from -1 to 1 , where 1 indicates identical vector directions, 0 indicates orthogonal vectors, and -1 indicates opposite directions. Thus, lower cosine similarity values

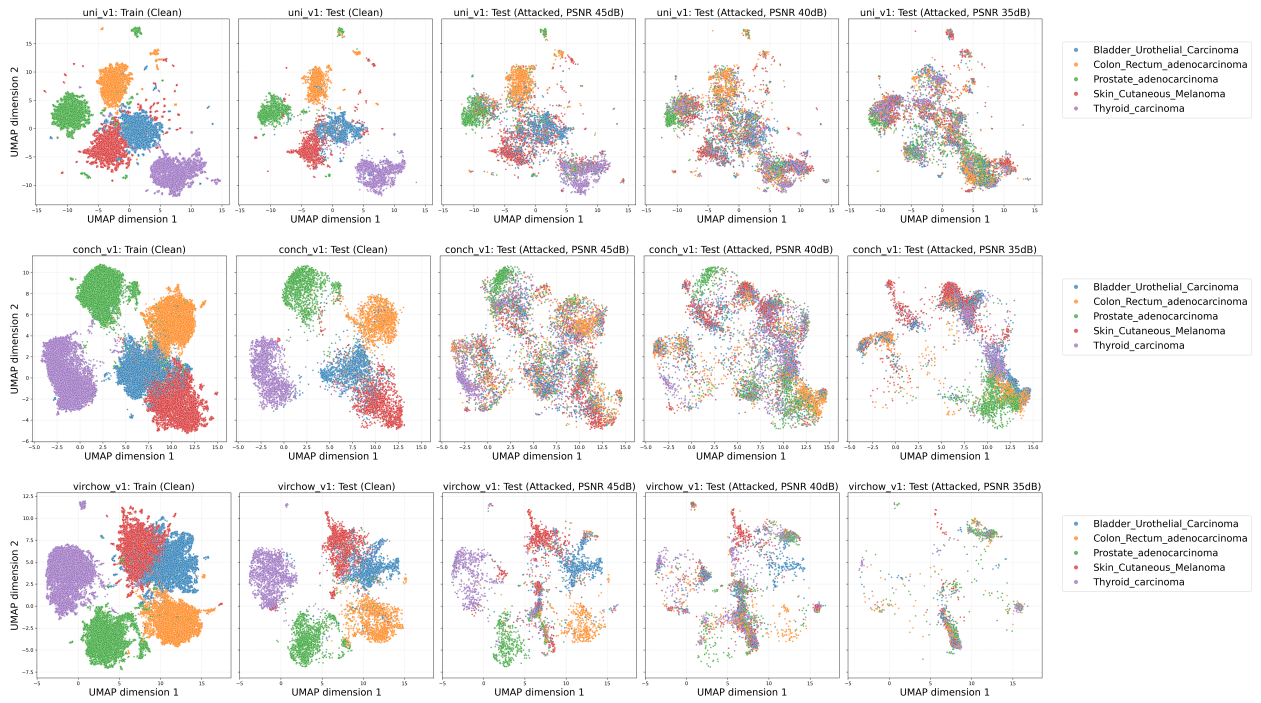


Figure 3: UMAP projections of the feature spaces of the PFMs in the clean scenario and under the white-box attack.

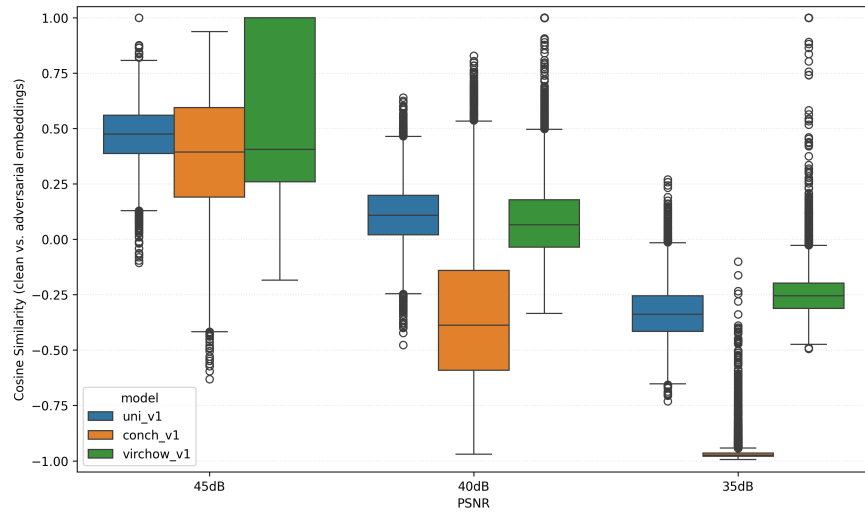


Figure 4: Cosine similarity between clean and adversarial embeddings at different PSNR values.

indicate a larger directional shift of the adversarial embedding relative to its clean counterpart. For each PFM, the response to the attack varies substantially across individual samples, with some embeddings shifting more than others at the same PSNR value. Despite this variability, a decrease in the median cosine similarity is observed as PSNR decreases. This trend is consistent with the optimization objective of the attack, which minimizes the cosine similarity between clean and adversarial embeddings. All PFMs exhibit relatively similar median cosine similarity values at 45 dB (0.39–0.47). However, as PSNR decreases, CONCH shows a substantially larger decrease in cosine similarity than UNI and Virchow. At 40 dB, CONCH already reaches a negative median cosine similarity (-0.39), while UNI and Virchow remain positive (0.11 and 0.07, respectively). The difference between CONCH and the other PFMs further increases at 35 dB, where CONCH reaches a median cosine similarity of -0.97 , compared with -0.34 for UNI and -0.25 for Virchow.

7.2 Attack Evaluation

Adversarial perturbations have a greater impact on linear probing performance in the white-box than in the black-box scenario, as shown in Figure 5.

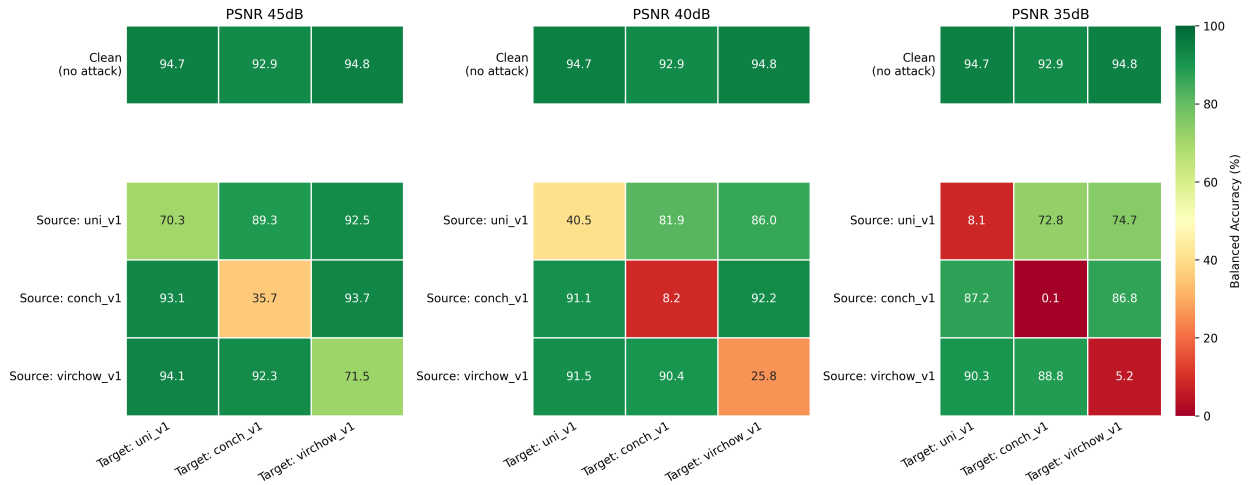


Figure 5: Linear probing performance under white-box and black-box attacks across PSNR values.

On the clean test set, all three linear probes achieve high balanced accuracy (94.7% for UNI, 92.9% for CONCH and 94.8% for Virchow).

In the white-box scenario (diagonal entries in the matrices), a consistent decrease in cancer-type discriminability is observed as PSNR decreases across all PFMs. Importantly, the classification layer is not directly attacked; rather, the adversarial perturbations applied to the input images alter the PFM feature representations, leading to a degradation in linear probing performance. At 45 dB, CONCH already exhibits a substantial drop in balanced accuracy, decreasing to 35.7%, compared with 70.3% for UNI and 71.5% for Virchow. As PSNR decreases, performance further deteriorates across all models, approaching a near-complete loss of linear cancer-type discriminability at 35 dB.

In contrast to the performance degradation observed in the white-box scenario, black-box attacks (off-diagonal entries in the matrices) result in smaller decreases in balanced accuracy across all PSNR values. Even at 35 dB, balanced accuracy remains substantially higher, ranging from 72.8% to 90.3%, than under white-box conditions.

7.3 Defense Evaluation

Gaussian smoothing partially mitigates the loss of cancer-type discriminability induced by the adversarial attack in the white-box scenario, as shown in Figure 6.

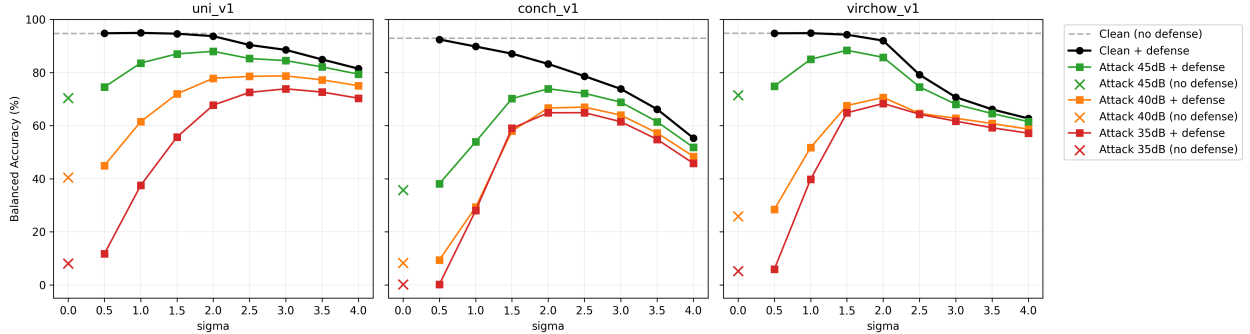


Figure 6: Linear probing performance on clean and adversarial images under Gaussian smoothing.

Even at 35 dB, Gaussian smoothing improves balanced accuracy, reaching peak values of 64.9% at $\sigma = 2.5$ for CONCH (compared with 0.1% without defense), 73.8% at $\sigma = 3$ for UNI (compared with 8.1%) and 68.3% at $\sigma = 2$ for Virchow (compared with 5.2%). However, across all PSNR values, the balanced accuracy on adversarial images with defense increases up to a peak at a certain value of σ , after which it no longer improves or progressively declines with further increases in σ . Since performance on the clean test set decreases as σ increases, this suggests that, beyond a certain value of σ , the benefit of attenuating the adversarial perturbation is outweighed by the loss of information relevant to cancer-type discriminability.

8 Ethical Implications

This study shows that the feature embeddings extracted by PFMs from histopathology patches are vulnerable to a task-agnostic adversarial attack. This raises concerns because a vulnerability in patch-level feature extraction may compromise the robustness of multiple downstream AI models operating at the WSI level, potentially posing a risk to patient safety and clinical trustworthiness. These considerations are particularly relevant in light of Article 15 of the EU AI Act, which requires high-risk AI systems to maintain appropriate levels of accuracy, robustness, and cybersecurity throughout their lifecycle. In this context, incorporating a PFM with vulnerable feature representations into a high-risk application may pose a challenge to meeting these requirements. The vulnerability also raises the question of how accountability should be shared between PFM developers and the teams that build downstream clinical applications on top of them. The vulnerability lies within the foundation model itself, but its potential consequences depend on how and where that model is subsequently used. This suggests that PFM developers should clearly communicate the limitations of the feature representations their models produce, while teams developing downstream clinical applications should assess whether the PFM is sufficiently robust for the intended application.

The findings of this study also highlight a tension between open science and AI security. Making PFMs openly accessible supports research and reproducibility, but also facilitates the development of white-box adversarial attacks by providing access to the model and its gradients. Such access enables the construction of targeted adversarial examples, which, as demonstrated in this study, substantially alter the feature representations of all three evaluated PFMs. This underscores the

challenge of balancing the benefits of open science with the need to mitigate the security risks associated with openly available AI models in healthcare.

9 Conclusion

This study evaluated the robustness of three state-of-the-art pathology foundation models — UNI, CONCH and Virchow — against a task-agnostic feature-space adversarial attack. The results show that all three PFMs are vulnerable to this type of attack under white-box conditions: UMAP visualization, embedding shift measured through cosine similarity, and linear probing all consistently reveal the vulnerability of the feature representations. Under black-box conditions, by contrast, the same attack does not disrupt the feature representations to the same extent. However, since this study evaluates PFM robustness against only one attack, these results cannot be taken to suggest that PFMs are less vulnerable to attacks crafted without access to the target model’s weights and gradients; indeed, the UTAP attack proposed by Wang et al., specifically designed to be universal and transferable, is shown to remain effective even under black-box conditions [15]. Gaussian smoothing partially mitigates this vulnerability, revealing a critical trade-off between the attenuation of adversarial perturbations and the loss of morphological details. Notably, the original publications introducing UNI, CONCH, and Virchow benchmark these models in terms of clean, non-adversarial performance across a range of downstream tasks, including patch-level linear probing as performed in this study. The present findings show that strong performance under clean conditions does not necessarily characterize the behaviour of the learned representations under adversarial perturbations. This suggests that adversarial robustness evaluation should be considered alongside standard performance benchmarking when assessing the suitability of PFMs for clinical use. Future work should extend this evaluation by quantifying how patch-level feature-space perturbations propagate to downstream WSI-level architectures to measure the actual impact on WSI-level diagnostic outputs. Additionally, future research should explore incorporating adversarial objectives directly into the PFM pre-training phase to provide downstream clinical applications with more robust and reliable feature representations.

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